10TH CLASS CHEMISTRY — NEW SYLLABUS 2025-26 (PECTAA)

Chapter 23 — Monohydroxy Alkanes or Alcohols

Solved Quick Checks + Exercise | New Book (PECTAA 2025-26)

Quick Checks (Solved)

23.1 What is the role of yeast in fermentation reaction?

Yeast contains enzymes which act as catalysts in the fermentation reaction. These enzymes break down glucose into ethanol and carbon dioxide, so without yeast the reaction cannot take place. Yeast is therefore added to the glucose solution to start and speed up the fermentation.

23.1 At which temperature enzymes present in yeast get inactive?

The temperature of the process must be kept around 35 °C to get the maximum yield. A lower temperature decreases the rate of the reaction, whereas a higher temperature makes the enzyme inactive. Therefore the yeast enzymes become inactive at high temperature, and yeast is also made inactive when the concentration of ethanol in the mixture exceeds 15 %, because ethanol is toxic to yeast.

23.2 Why oxygen is excluded when the fermentation reaction is carried out?

Fermentation must be carried out in the absence of air (anaerobic conditions) to obtain ethanol, because oxygen converts the resulting ethanol into ethanoic acid. If air is present, the yield of ethanol decreases and the product becomes impure.

23.2 Which method for the preparation of ethene involves renewable source?

Fermentation involves a renewable source. Ethanol is prepared by the fermentation of glucose obtained from cane-sugar and other plant materials, and plants can be grown again and again. (The book prints the word “ethene” here, but the method being compared is the preparation of **ethanol**; fermentation gives ethanol by a renewable route, while hydration of ethene uses crude oil, which is non-renewable.)

23.3 Why does butanol evolve a higher amount of heat than propanol when burnt in oxygen?

Butanol has one more $–CH_2$ group than propanol, so one mole of butanol contains more carbon and hydrogen atoms. On complete combustion it produces more CO_2 and H_2O per mole, and burning of the extra C–H bonds releases extra heat. Therefore the heat evolved by butanol (about 2676 kJ per mole) is higher than that of propanol (about 2017 kJ per mole).

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23.4 Why does the amount of heat evolved during burning of alcohols go on increasing from lower to higher alcohols?

During combustion of alcohols, the weaker C–H bonds in the $–CH_2$ groups break and the stronger C=O and O–H bonds are formed in CO_2 and H_2O respectively. Each higher alcohol has one more $–CH_2$ group than the previous one, so more C–H bonds are broken and more CO_2 and H_2O are formed per mole. Hence the amount of heat evolved increases with the addition of a $–CH_2$ group in the homologous series.

23.4 Why burning of alcohols does not produce oxides of sulphur unlike fossil fuels?

Alcohols contain only carbon, hydrogen and oxygen. They contain no sulphur at all, so their complete combustion gives only carbon dioxide and water. Fossil fuels contain sulphur as an impurity, therefore burning them produces sulphur dioxide and sulphur trioxide, which cause acid rain and air pollution. For this reason alcohols are considered environment friendly fuels.

Exercise (Solved)

A. Multiple Choice Questions

1. Which conditions are considered best for the fermentation of glucose?

(a) 35 °C, fresh yeast, absence of oxygen
(b) 45 °C, yeast
(c) 45 °C, absence of oxygen
(d) 35 °C, fresh yeast

Answer: (a) — Glucose is converted to ethanol at around 35 °C in the presence of yeast and in the absence of air (anaerobic conditions). At 45 °C the enzymes become inactive, and if oxygen is present the ethanol is oxidised to ethanoic acid.

2. Which catalyst other than H_3PO_4 can also be used for hydration of alkenes?

(a) NaOH
(b) H_2SO_4
(c) CH_3COOH
(d) Ni

Answer: (b) — Hydration of an alkene needs an acid catalyst, and sulphuric acid also supplies H^+ ions for the addition of water across the double bond. Sodium hydroxide is a base, ethanoic acid is a weak organic acid and nickel is used for hydrogenation, not hydration.

3. Alcohol which has the maximum heat of combustion is:

(a) $CH_3CH_2CH_2OH$
(b) CH_3CH_2OH
(c) CH_3OH
(d) $CH_3CH_2CH_2CH_2OH$

Answer: (d) — Heat of combustion increases with the size of the carbon chain, because each extra $–CH_2$ group adds more C–H bonds which burn and release heat. Butan-1-ol has four carbon atoms, so it gives the maximum heat (about 2676 kJ per mole).

4. The most environment friendly fuel is:

(a) coal (b) kerosene oil

(c) ethanol (d) wood

Answer: (c) — Ethanol contains no sulphur, so on burning it does not produce oxides of sulphur which cause acid rain. It can also be prepared from renewable plant material, so it is the most environment friendly fuel among these.

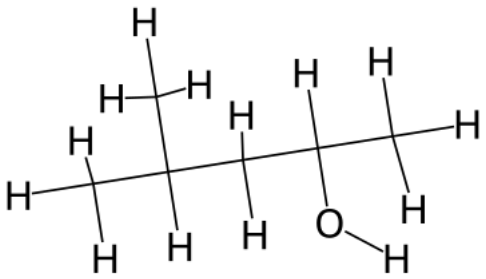
5. Which property or properties of ethanol make it suitable to be used in cosmetic industry?

(a) Easily flammable (b) Highly volatile and deodorizing

(c) Antipyretic (d) Sedative

Answer: (b) — Ethanol is highly volatile, so it evaporates in a very short time, and it also has deodorizing and anti-inflammatory properties. These properties make it suitable for perfumes, deodorants and after shave lotions.

6. Which is the correct name of the following alcohol:
CH_3–CH(OH)–CH_2–CH(CH_3)–CH_3



(a) 2-Methylpentan-4-ol (b) 4-Methylpentan-2-ol

(c) Hexan-2-ol (d) Hexan-4-ol

Answer: (b) — The longest chain contains five carbon atoms (pentan-), the –OH group is on carbon 2 and the methyl group is on carbon 4, so the name is 4-methylpentan-2-ol.

7. Which property of ethanol makes it suitable for cleaning wounds?

(a) Good solvent (b) Volatility

(c) Antiseptic (d) Deodorizing

Answer: (c) — Ethanol has bactericidal (antiseptic) action, so it kills the germs present in a wound. That is why it is used as a disinfectant in methylated spirit and hand sanitizer.

8. Which alcohol is used to accomplish all the following tasks? It is added in petrol to improve combustion. It serves as a feed stock to prepare vinegar and many other organic compounds.

(a) Propanol (b) Propan-2-ol

(c) Ethanol (d) Methanol

Answer: (c) — Ethanol is blended with petrol to improve combustion, and it is oxidised to ethanoic acid (vinegar) and used as a feedstock for many organic compounds.

B. Short Answer Questions

23.1 Give two advantages of burning of alcohols over the burning of fossil fuel?

(i) Alcohols are obtained from renewable sources, because they are produced from plant materials, while fossil fuels are non-renewable.

(ii) Burning of alcohols does not produce oxides of sulphur, so alcohols are environment friendly as compared to fossil fuels. They also have higher fuel efficiency as compared to petrol or diesel.

23.2 Which properties of ethyl alcohol make it useful for cosmetic industry?

Ethyl alcohol is highly volatile, so it evaporates in a very short time. It has deodorizing, anti-inflammatory, refreshing and antimicrobial properties. It is also a good solvent for perfumes and other organic compounds which are insoluble in water. Due to these properties it is used in hair styling products, foundation creams, perfumes, deodorants and after shave lotions.

23.3 How is ethyl alcohol important for pharmaceutical industry?

Ethyl alcohol possesses bactericidal activity, so it is often used as a disinfectant in hand sanitizer and methylated spirit. It is also commonly used as a solvent and preservative in many pharmaceutical preparations, medicines, tinctures and plant extracts.

23.4 Give any two disadvantages of fermentation reactions.

(i) The rate of reaction is very low; it may take days to complete, and the reaction has to be carried out in the absence of oxygen.

(ii) The reaction produces only about 15 % alcohol which needs further distillation to increase the concentration, and distillation requires energy.

23.5 Write the conditions involved in the hydration of ethene.

A mixture of ethene and steam is passed over hot phosphoric acid (H_3PO_4), which acts as a catalyst, at about 300 °C and 60 atm pressure. Water is added across the double bond of ethene and ethanol is formed. The gaseous ethanol obtained is then condensed into its liquid form.

$$CH_2{=}CH_{2(g)} + H_2O_{(g)} \rightarrow CH_3CH_2OH_{(g)} \quad (H_3PO_4,\ 300\ °C,\ 60\ atm)$$

C. Constructed Response Questions

23.1 How ethanol is obtained as a by-product in sugar industry?

In the sugar industry, sugar is crystallised from cane juice and a thick dark liquid called molasses is left behind. Molasses still contains a good amount of sugar which cannot be crystallised further. This molasses is diluted with water, mixed with yeast and kept at around 35 °C in the absence of oxygen. The enzymes present in yeast ferment the sugar of molasses into ethanol and carbon dioxide. The dilute ethanol thus obtained is then separated and concentrated by fractional distillation. In this way ethanol is obtained as a by-product of the sugar industry.

$$C_6H_{12}O_6 \rightarrow 2C_2H_5OH + 2CO_2 \quad \text{(yeast, 35 °C)}$$

23.2 Why do we get relatively impure alcohol in the fermentation process?

Fermentation takes place in water and uses natural raw materials such as molasses, so the product is a mixture of water, unused sugar, yeast cells and small amounts of other products. Moreover, yeast becomes inactive when the concentration of ethanol exceeds 15 %, because ethanol is toxic to yeast, so the fermentation stops at a dilute stage. The alcohol obtained is therefore only about 15 % and is impure; it must be separated from water and other impurities by fractional distillation to get pure ethanol.

23.3 How do properties of ethyl alcohol help us to utilize it as fuel?

Ethyl alcohol is flammable and burns in air to produce carbon dioxide and water, evolving a large amount of heat, so it can be used as a fuel in internal combustion engines. It contains oxygen in its molecule, so it burns completely with little soot when sufficient air is supplied. It does not contain sulphur, so it does not produce oxides of sulphur which pollute the air. It is completely miscible with petrol, so it can be blended with petrol to improve combustion, and it can be prepared from renewable plant material.

$$C_2H_5OH + 3O_2 \rightarrow 2CO_2 + 3H_2O \quad \Delta H = -1367 \text{ kJ/mol}$$

D. Descriptive Questions

23.1 Give a comparison of fermentation reaction and catalytic hydration of ethene for the preparation of ethanol.

Fermentation	Catalytic hydration of ethene
The reaction takes place at around 35 °C and at atmospheric pressure.	The reaction is performed at high temperature (about 300 °C) and at high pressure (about 60 atm).
The starting material is a renewable source such as cane-sugar.	The starting material (ethene) is obtained from crude oil, which is a non-renewable source.
The rate of reaction is low; it may take days to complete.	The rate of reaction is high and it can be run continuously.
It produces about 15 % alcohol which needs further distillation to increase the concentration, and distillation requires energy.	It produces relatively pure alcohol.
The ethanol produced by this process is cheap.	The ethanol produced by this method is expensive.

$$C_6H_{12}O_6 \rightarrow 2C_2H_5OH + 2CO_2 \quad \text{(yeast, 35 °C)}$$

$$CH_2{=}CH_2 + H_2O \rightarrow C_2H_5OH \quad (H_3PO_4, 300\ °C, 60\ atm)$$

23.2 Describe the preparation of ethanol by fermentation process.

A dilute sugar solution, usually obtained from molasses, is taken in a vessel and mixed with fresh yeast. The mixture is kept at around 35 °C in a neutral or slightly acidic solution, and air is excluded so that anaerobic conditions are maintained. The enzymes present in yeast act as catalysts and convert glucose into ethanol and carbon dioxide.

$$C_6H_{12}O_6 \rightarrow 2C_2H_5OH + 2CO_2 \quad \text{(yeast, 35 °C)}$$

Carbon dioxide gas escapes from the mixture, leaving behind a dilute solution of ethanol in water. The temperature must be kept around 35 °C: a lower temperature decreases the rate of the reaction, while a higher temperature makes the enzyme inactive. Oxygen must also be excluded because it converts the ethanol into ethanoic acid. Since ethanol is toxic to yeast in high concentration, the fermentation stops when the concentration of ethanol exceeds about 15 %. The ethanol is then separated from yeast and water by fractional distillation, and in this way ethanol of high concentration is obtained.

23.3 Describe the applications of ethanol in pharmaceutical and cosmetic industries?

Pharmaceutical industry:

1. Ethyl alcohol possesses bactericidal activity, so it is used as a disinfectant in hand sanitizer and methylated spirit.

2. It is commonly used as a solvent and preservative in many pharmaceutical preparations, so it dissolves medicines, tinctures and plant extracts.

3. Being an antiseptic, it is used for cleaning wounds and for dressing them, and methylated spirit is kept in the first aid kit of every home.

Cosmetic industry:

1. Being highly volatile, ethanol evaporates in a very short time, so it helps the cosmetic product to dry quickly after application.

2. It has deodorizing, anti-inflammatory, refreshing and antimicrobial properties, so it is used in deodorants and after shave lotions.

3. It is a good solvent for perfumes and other organic compounds which are insoluble in water, so it is an important part of hair styling products, foundation creams, perfumes and lotions.

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